

Lecture 22/Module 3

Sinusoidal Steady State Analysis

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Today's Topics

- Phasors
- Phasor Relationships for Circuit Elements

Sinusoids and Phasors

Sinusoids are easily expressed in terms of **phasors**, which are more convenient to work with than sine and cosine functions.

A phasor is a complex number that represents the amplitude and phase of a sinusoid.

Phasors provide a simple means of analyzing linear circuits excited by sinusoidal sources.

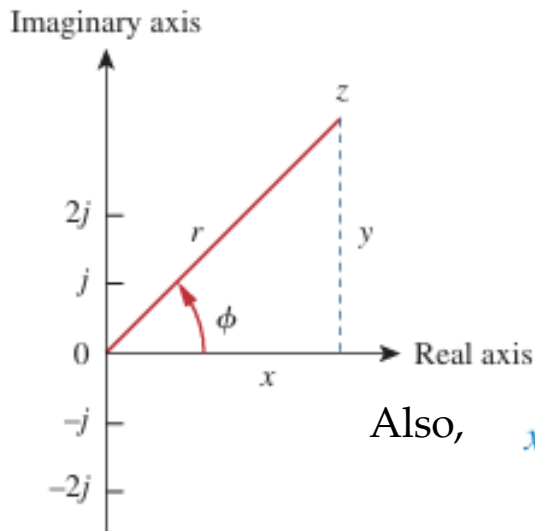
It can be represented in anyone of the 3 forms.

$$z = x + jy \quad \text{Rectangular form}$$

$$z = r \angle \phi \quad \text{Polar form}$$

$$z = re^{j\phi} \quad \text{Exponential form}$$

where r is the magnitude of z , and ϕ is the phase of z .



X axis represents the real part and the y axis represents the imaginary part of a complex number.

$$r = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \frac{y}{x}$$

The relationship between the rectangular form and the polar form.

$$x + jy = r \angle \phi.$$

Also, $x = r \cos \phi, \quad y = r \sin \phi$

$$z = x + jy = r \angle \phi = r(\cos \phi + j \sin \phi) \quad 3$$

Sinusoids and Phasors

Addition:

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

Subtraction:

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

Multiplication:

$$z_1 z_2 = r_1 r_2 \angle \phi_1 + \phi_2$$

Division:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle \phi_1 - \phi_2$$

Reciprocal:

$$\frac{1}{z} = \frac{1}{r} \angle -\phi$$

Square Root:

$$\sqrt{z} = \sqrt{r} \angle \phi/2$$

Complex Conjugate:

$$z^* = x - jy = r \angle -\phi = r e^{-j\phi}$$

Note that from Eq. (9.18e),

$$\frac{1}{j} = -j$$

Addition and subtraction of complex numbers are better performed in rectangular form.

Multiplication and division are better done in polar form.

The idea of phasor representation is based on Euler's identity.

$$e^{\pm j\phi} = \cos \phi \pm j \sin \phi$$

$$\cos \phi = \operatorname{Re}(e^{j\phi})$$

$$\sin \phi = \operatorname{Im}(e^{j\phi})$$

Sinusoids and Phasors

$$v(t) = V_m \cos(\omega t + \phi) = \operatorname{Re}(V_m e^{j(\omega t + \phi)})$$

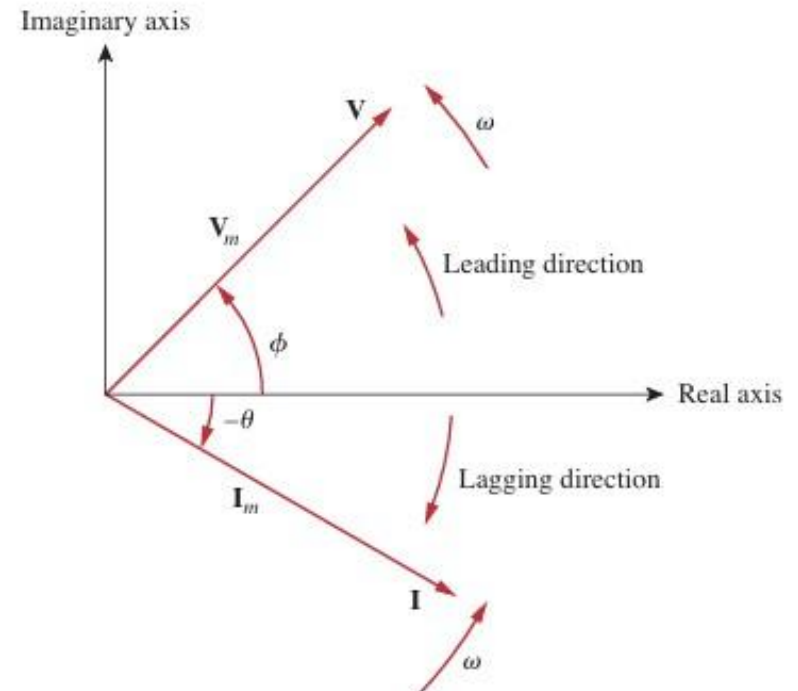
$$v(t) = \operatorname{Re}(V_m e^{j\phi} e^{j\omega t})$$

$$v(t) = \operatorname{Re}(\mathbf{V} e^{j\omega t})$$

Where $\mathbf{V}$ is the phasor representation of the sinusoid $v(t)$.

$$\mathbf{V} = V_m e^{j\phi} = V_m \angle \phi$$

phasor diagram



$$\mathbf{V} = V_m \angle \phi \text{ and } \mathbf{I} = I_m \angle -\theta.$$

Sinusoids and Phasors

Domain Transformations:

$$\begin{array}{ccc} v(t) = V_m \cos(\omega t + \phi) & \Leftrightarrow & \mathbf{V} = V_m \angle \phi \\ \text{(Time-domain} & & \text{(Phasor-domain} \\ \text{representation)} & & \text{representation)} \end{array}$$

Sinusoid-phasor transformation.

Time domain representation

Phasor domain representation

$$V_m \cos(\omega t + \phi)$$

$$V_m \angle \phi$$

$$V_m \sin(\omega t + \phi)$$

$$V_m \angle \phi - 90^\circ$$

$$I_m \cos(\omega t + \theta)$$

$$I_m \angle \theta$$

$$I_m \sin(\omega t + \theta)$$

$$I_m \angle \theta - 90^\circ$$

Sinusoids and Phasors

$$\frac{dv}{dt} \quad \Leftrightarrow \quad j\omega V$$

(Time domain) (Phasor domain)

Differentiating a sinusoid is equivalent to multiplying its corresponding phasor by $j\omega$.

$$\int v \, dt \quad \Leftrightarrow \quad \frac{V}{j\omega}$$

(Time domain) (Phasor domain)

Integrating a sinusoid is equivalent to dividing its corresponding phasor by $j\omega$.

1. $v(t)$ is the instantaneous or time domain representation, while $\mathbf{V}$ is the frequency or phasor domain representation.
2. $v(t)$ is time dependent, while $\mathbf{V}$ is not.
3. $v(t)$ is always real with no complex term, while $\mathbf{V}$ is generally complex.

Phasor analysis applies only when frequency is constant; it applies in manipulating two or more sinusoidal signals only if they are of the same frequency.

Sinusoids and Phasors

Evaluate:

$$(40 \angle 50^\circ + 20 \angle -30^\circ)$$

Transform these sinusoids to phasors:

(a) $i = 6 \cos(50t - 40^\circ)$ A

(b) $v = -4 \sin(30t + 50^\circ)$ V

Find the sinusoids represented by these phasors:

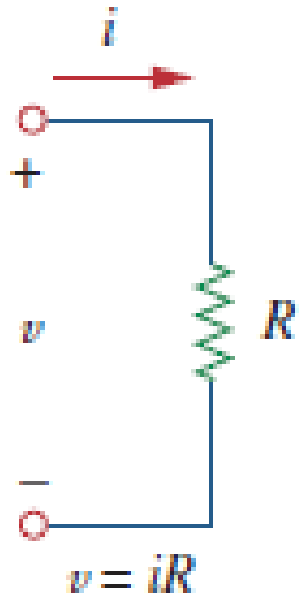
(a) $\mathbf{I} = -3 + j4$ A

(b) $\mathbf{V} = j8e^{-j20^\circ}$ V

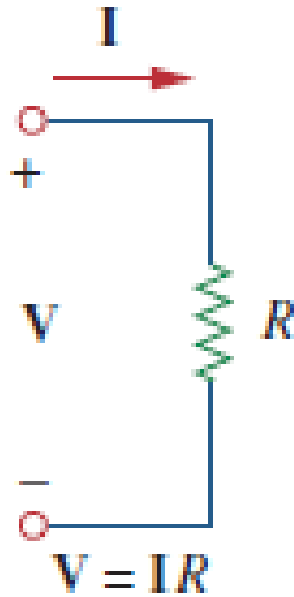
Given $i_1(t) = 4 \cos(\omega t + 30^\circ)$ A and $i_2(t) = 5 \sin(\omega t - 20^\circ)$ A, find their sum.

Phasor relationship for circuit elements

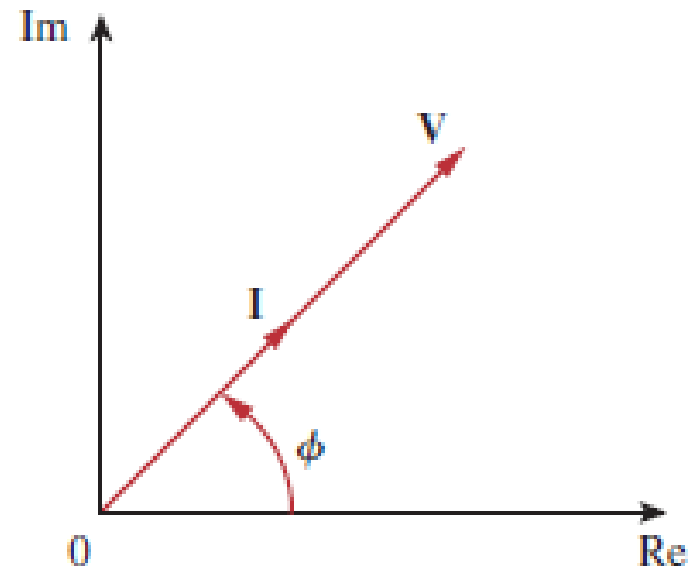
RESISTOR



(a)



(b)



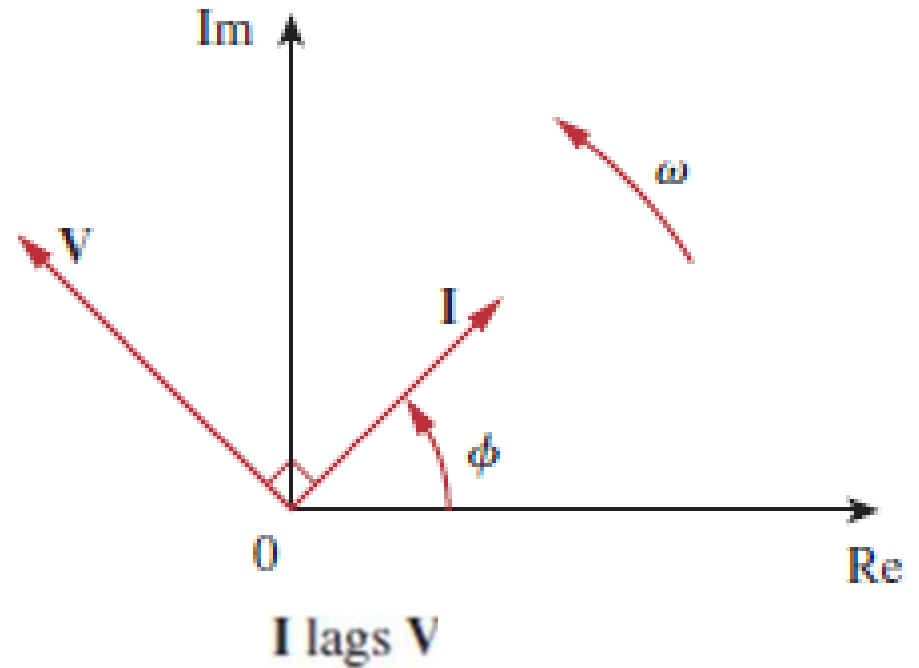
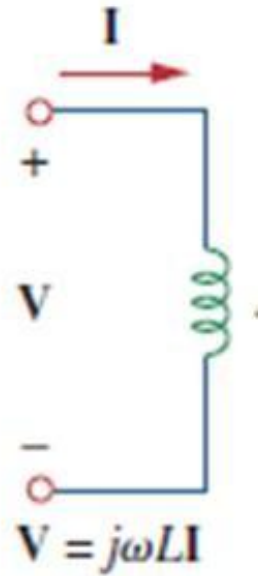
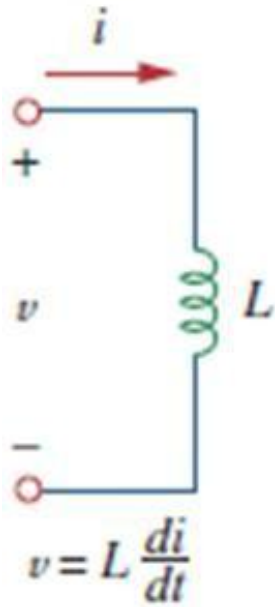
$$v = iR = RI_m \cos(\omega t + \phi)$$

$$V = RI_m \angle \phi$$

$$V = RI$$

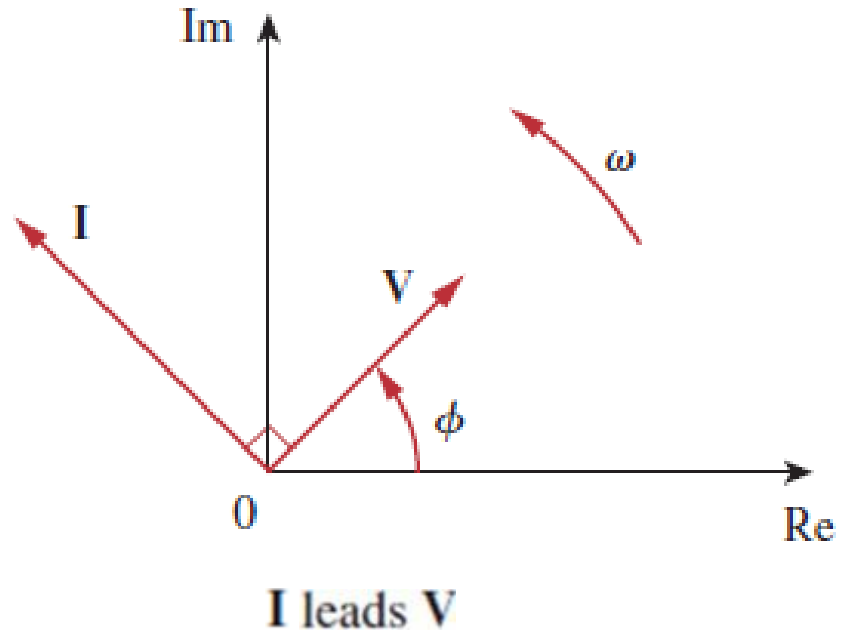
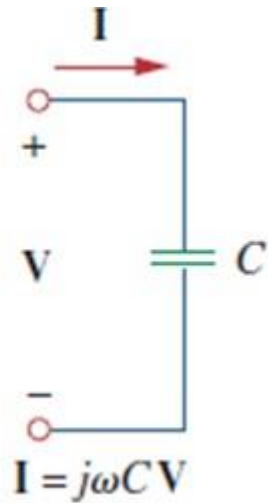
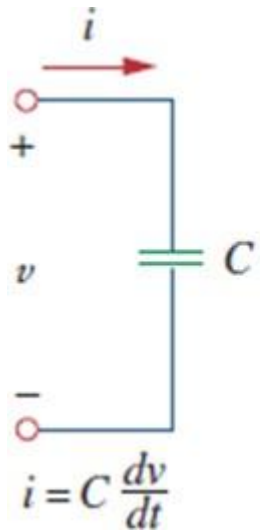
Phasor relationship for circuit elements

INDUCTOR



Phasor relationship for circuit elements

CAPACITOR



Phasor relationship for circuit elements

Voltage – Current relationship in circuit elements

Element	Time domain	Frequency domain
R	$v = Ri$	$V = RI$
L	$v = L \frac{di}{dt}$	$V = j\omega LI$
C	$i = C \frac{dv}{dt}$	$V = \frac{I}{j\omega C}$

Thank You